MICROSOFT POWER PLATFORM PROJECT

Employee Wellness Management System

Project Documentation

Item	Details
Solution name	Employee Wellness Solution
Applications	Employee Fitness (Canvas app) | Employee Wellness Management Admin (Model-driven app)
Technologies	Power Apps | Microsoft Dataverse | Power Automate | Power BI
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1. Problem Statement

Organisations that want to encourage healthier employees often track wellness activity in spreadsheets, chat messages or on paper. This makes the information scattered, inconsistent and difficult to act on.

The key problems this project addresses are:

- **No central record.** Employee wellness details and activities live in disconnected files, so HR has no single source of truth.
- **No simple way for employees to log activities.** Employees need a quick, mobile-friendly way to record what they plan to do and what they have completed.
- **No controlled activity status.** Without a defined lifecycle (Pending, Completed, Cancelled) it is impossible to measure real participation.
- **Poor data quality.** Free-form entry leads to invalid dates, missing values and duplicate or inconsistent data.
- **No reminders or confirmations.** Employees forget planned activities because nothing prompts them, and nothing confirms that an entry was saved.
- **Limited visibility for management.** HR cannot easily see participation, calories burned or wellness points across employees.

> **Problem in one line**
>
> The organisation needs one governed platform where employees log and complete wellness activities, HR administers the data, and routine communication is automated.

2. Use Case

2.1 Actors

Actor	Role in the solution	Application used
Employee	Views their profile, logs wellness activities, completes or cancels pending activities and tracks personal progress.	Employee Fitness (Canvas app)
HR / Wellness Coordinator	Maintains employee wellness profiles, reviews activity records and monitors participation.	Employee Wellness Management Admin (Model-driven app)
Administrator	Configures Dataverse tables, apps, flows, business rules and security.	Power Apps maker portal, Power Automate

2.2 Primary Use Cases

#	Use case	Description
UC-1	View my profile	Employee opens My Profile to see department, designation, BMI, activity level, wellness score, goal and enrollment date.
UC-2	Add a wellness activity	Employee submits a new activity with type, date, duration, calories, points and remarks. The status is set to Pending.
UC-3	View my activities	Employee sees a list of their activities with date, duration and status.
UC-4	Complete an activity	Employee taps Mark Complete on a pending activity; the status becomes Completed.
UC-5	Cancel an activity	Employee taps Cancel on a pending activity; the status becomes Cancelled.
UC-6	Track my progress	Employee views totals for completed activities, points, minutes, calories and a progress percentage.
UC-7	Receive notifications	Employee gets an email when an activity is added or completed, and reminders for pending activities.

#	Use case	Description
UC-8	Administer wellness data	HR creates and edits profiles and activities through views and forms in the Model-driven app.

2.3 End-to-End User Journey

1. An employee profile is created and the Enrollment Date is populated automatically.
2. The employee opens the Employee Fitness app and lands on the Home screen.
3. The employee opens Add Activity, enters the details and selects Submit.
4. Dataverse creates the Wellness Activity with Participation Status = Pending.
5. A confirmation email is sent by Power Automate.
6. A reminder is sent while the activity is still pending.
7. The employee performs the activity and selects Mark Complete.
8. The My Progress screen updates, and the data is available for reporting in Power BI.

3. Project Overview

The Employee Wellness Management System is a Microsoft Power Platform solution that helps an organisation record, manage, automate and analyse employee wellness activities. Microsoft Dataverse is the central data platform; a Canvas app serves employees, a Model-driven app serves administrators, Power Automate handles notifications and background processing, and Power BI provides the analytics layer.

3.1 Purpose

The purpose of the project is to replace manual, disconnected tracking with a single, secure and automated wellness platform that is simple for employees to use and easy for HR to manage and extend.

3.2 Objectives

- Create a central Dataverse data model for employee wellness information.
- Build an employee-friendly, mobile-style Canvas app for entering and tracking activities.
- Build a Model-driven app for structured administration and data management.
- Automate confirmations, reminders and other activity-related notifications with Power Automate.
- Apply business rules and validation to improve data quality.
- Provide a foundation for Power BI dashboards on participation and wellness trends.
- Deliver a solution that can be extended with additional wellness features.

3.3 Scope

In scope

- Dataverse tables: Employee Wellness Profile and WellnessActivity, with choice, lookup and autonumber columns.
- Canvas app "Employee Fitness" with Home, My Profile, Activities, Add Activity and My Progress screens.
- Model-driven app "Employee Wellness Management Admin" with views and forms for both tables.
- Four Power Automate cloud flows for notifications, reminders and profile automation.
- Business rule for Activity Date validation.
- Reporting design for Power BI on top of the wellness data.

Out of scope (future enhancements)

- Integration with wearable devices or external fitness trackers.
- Team challenges, leaderboards and rewards.
- Mobile push notifications and Microsoft Teams adaptive cards.

4. Solution Architecture and Components

The solution follows a layered architecture in which Dataverse is the single source of truth. Both apps read and write the same tables, Power Automate reacts to data changes, and Power BI consumes the data for reporting.

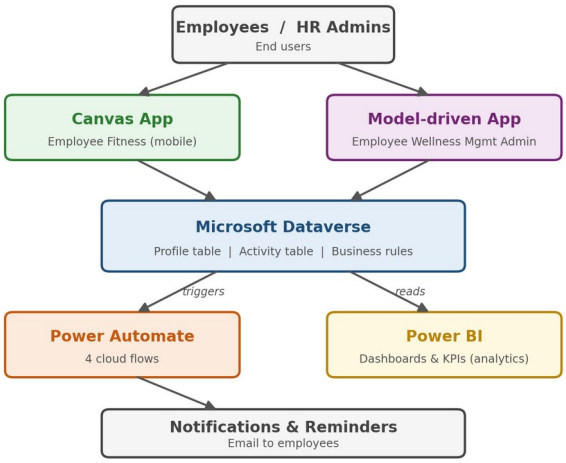


Figure 1: Solution architecture

4.1 Components

Component	Name	Responsibility
Solution	Employee Wellness Solution	Packages all tables, apps, flows and rules for transport between environments.
Data platform	Microsoft Dataverse	Stores profiles and activities; provides choice columns, lookups, autonumber, business rules and security.
Canvas app	Employee Fitness	Employee-facing phone-style app for profile, activities, activity entry and progress.
Model-driven app	Employee Wellness Management Admin	Administrator interface with views and forms for both tables.
Automation	4 cloud flows	Confirmations, reminders, completion handling and new-profile automation.
Business logic	BR - Activity Date Validation	Dataverse business rule that keeps Activity Date valid.
Analytics	Power BI	Dashboards for participation, calories, points and trends.

4.2 Design Principles

Dataverse is the single source of truth and is never duplicated inside the apps. Employees and administrators each get the app that suits them, routine communication is automated with Power Automate, and data quality is enforced at the source through choice columns, lookups and business rules.

5. Data Model

Dataverse was chosen as the central data store because it supports structured business data, relationships, Choice and Lookup columns, security, business rules and native integration with Power Apps and Power Automate. The data model contains two custom tables in the Employee Wellness Solution, using the publisher prefix pj_.

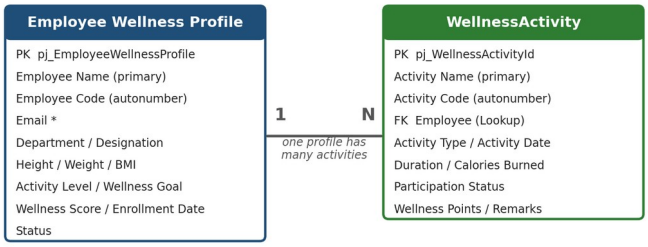


Figure 2: Table relationship (Employee Wellness Profile 1 : N WellnessActivity)

5.1 Employee Wellness Profile

Stores the employee information used by the wellness solution. The primary name column is Employee Name.

Display name	Schema name	Data type	Required	Purpose
Employee Name	pj_EmployeeName	Single line of text (primary)	Yes	Full name of the employee
Employee Code	pj_EmployeeCode	Autonumber	No	Auto-generated business identifier (e.g. Emp-10001)
Email	pj_Email	Email	Yes	Employee email address
Department	pj_Department	Choice	No	Employee department (e.g. HR, Finance, IT, Marketing)
Designation	pj_Designation	Single line of text	No	Job title
Enrollment Date	pj_EnrollmentDate	Date only	No	Date the profile was enrolled; populated automatically
Status	pj_Status	Choice	No	Profile status (e.g. Active)
Height (in meter)	pj_Heightinmeter	Decimal	No	Height used for BMI
Weight (kg)	pj_Weightkg	Decimal	No	Weight used for BMI
BMI	pj_BodyMassIndex	Decimal (calculated)	No	Body Mass Index calculated from height and weight
Activity Level	pj_ActivityLevel	Choice	No	General activity level (e.g. Moderate)
Wellness Goal	pj_WellnessGoal	Choice	No	Goal such as Weight Loss, Stress Reduction, Fitness Improvement
Wellness Score	pj_WellnessScore	Decimal	No	Overall wellness score
Employee Wellness Profile	pj_EmployeeWellnessProfileId	Unique identifier	Yes	System primary key

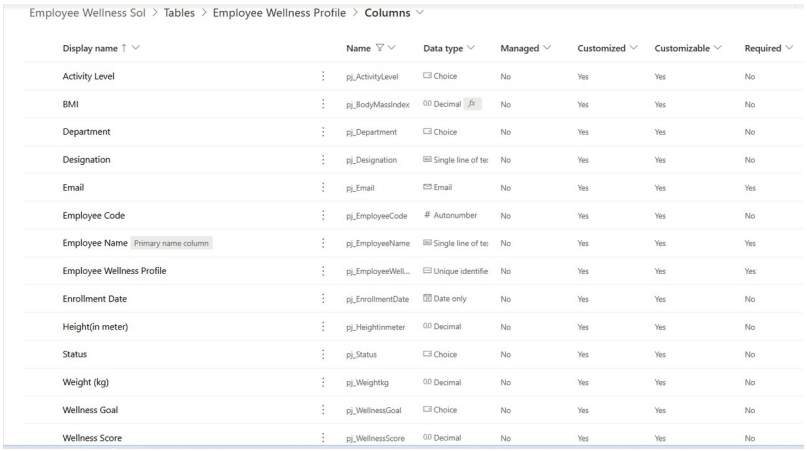


Figure 3: Employee Wellness Profile columns in the maker portal

5.2 WellnessActivity

Stores each wellness activity created by or for an employee. The primary name column is Activity Name.

Display name	Schema name	Data type	Purpose
Activity Name	pj_ActivityName	Single line of text (primary)	Name of the activity
Activity Code	pj_ActivityCode	Autonumber	Unique reference in A-0001 style
Activity Type	pj_ActivityType	Choice	Category: Exercise, Meditation, Nutrition, Sports, Health
Activity Date	pj_ActivityDate	Date only	Date the activity is planned or performed
Duration (Minutes)	pj_DurationMinutes	Whole number	Length of the activity
Calories Burned	pj_CaloriesBurned	Whole number	Calories associated with the activity
Employee	pj_Employee	Lookup	Links the activity to an Employee Wellness Profile
Participation Status	pj_ParticipationStatus	Choice	Lifecycle status: Pending, Completed, Cancelled
Wellness Points	pj_WellnessPoints	Whole number	Points earned for the activity
Remarks	pj_Remarks	Multiple lines of text	Additional notes
WellnessActivity	pj_WellnessActivityId	Unique identifier	System primary key

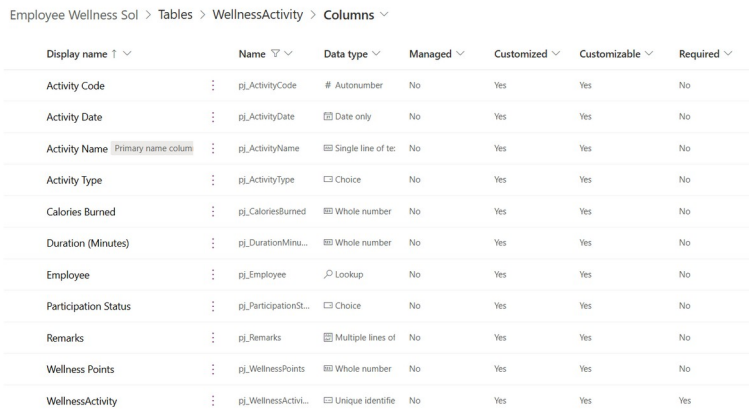


Figure 4: WellnessActivity columns in the maker portal

5.3 Choice Columns

Column	Values
Participation Status	Completed, Pending, Cancelled
Activity Type	Exercise, Meditation, Nutrition, Sports, Health
Department	HR, Finance, IT, Marketing and other departments
Wellness Goal	Weight Loss, Stress Reduction, Fitness Improvement
Activity Level	Moderate and other activity levels

Choice values are stored as numbers in Dataverse; both apps display the human-readable labels.

5.4 Relationship

WellnessActivity has a many-to-one relationship to Employee Wellness Profile through the Employee lookup column (pj_Employee). One employee profile can therefore have many wellness activities. The Canvas app shows the employee name to the user while Dataverse stores the relationship using the record's unique identifier.

5.5 Autonumber Columns

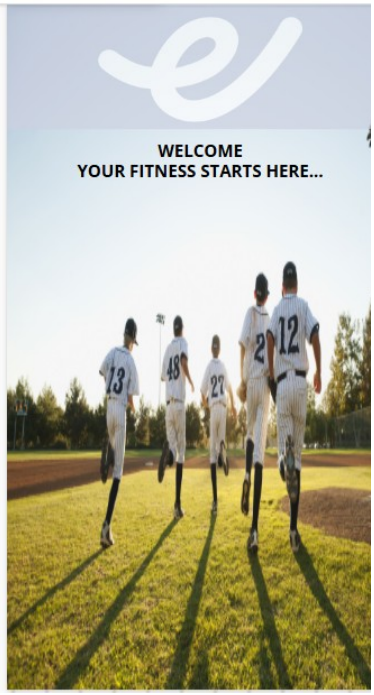
- Employee Code identifies the employee for business use instead of a raw Employee ID.

- Activity Code gives each activity a business-friendly reference in the format A-0001, A-0002, A-0003, ...

6. Canvas App: Screens and Functionality

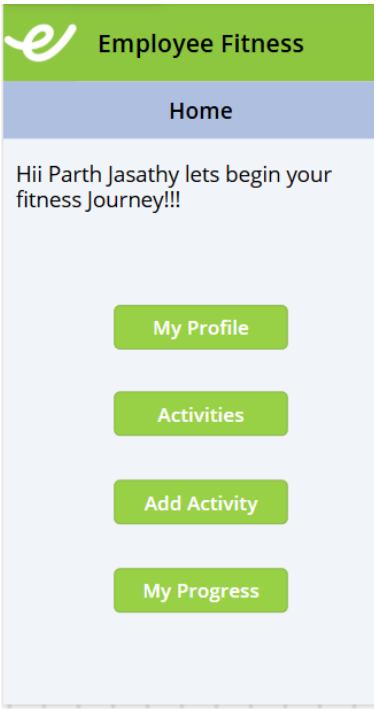
Employee Fitness is the employee-facing Canvas app. It is designed for a phone-style layout with a green header bar, a blue-grey title strip for each screen, and large green action buttons. Every screen other than Home has a back arrow in the header.

6.1 Screens



Welcome Screen

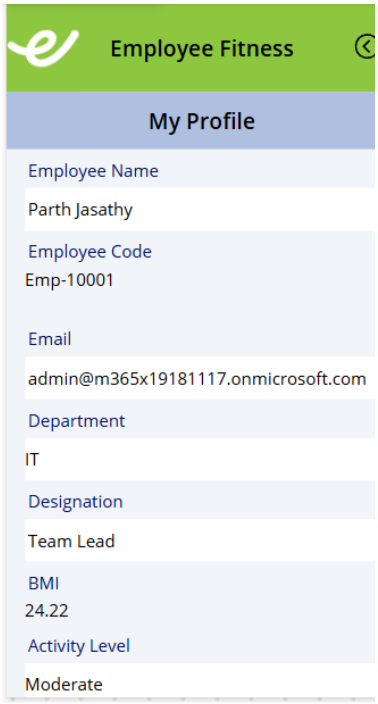
Branded splash screen with the logo and the message "Welcome - Your fitness starts here...".



Home Screen

Greets the signed-in employee by name and provides navigation to the four main areas of the app.

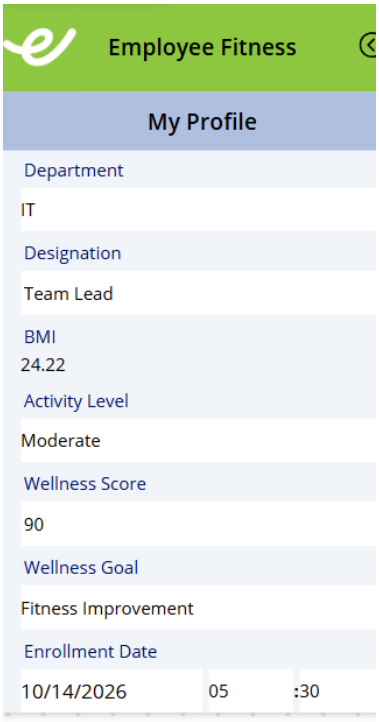
- **My Profile** - opens the employee's wellness profile.
- **Activities** - lists all of the employee's activities.
- **Add Activity** - opens the new activity form.
- **My Progress** - shows wellness progress.



My Profile (top)

Displays the employee's profile record from Dataverse. The first part of the screen shows:

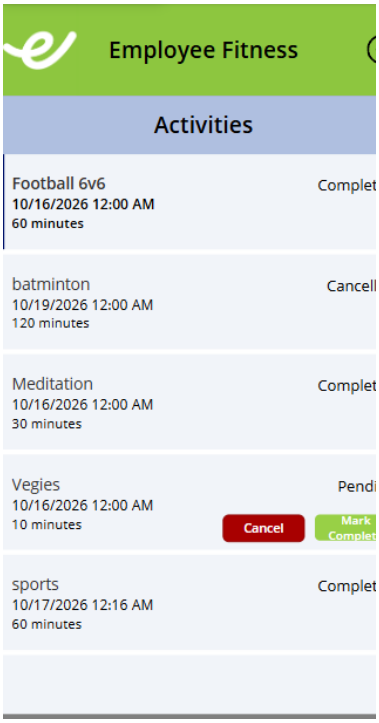
- Employee Name and Employee Code (for example Emp-10001)
- Email, Department and Designation
- BMI (24.22) and Activity Level (Moderate)



My Profile (bottom)

Scrolling down shows the remaining wellness details:

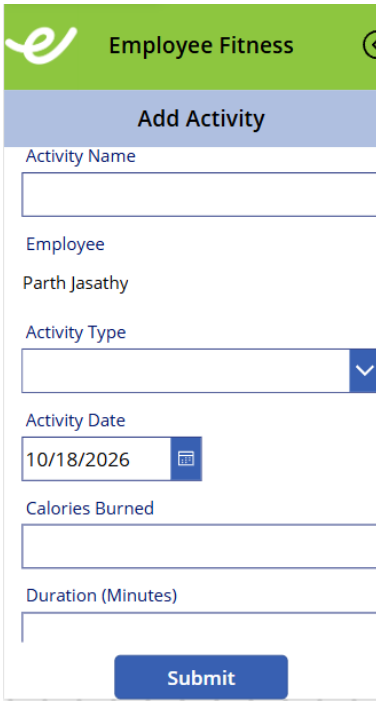
- Wellness Score (90) and Wellness Goal (Fitness Improvement)
- Enrollment Date, which is set automatically when the profile is created



Activities Screen

A vertical gallery lists the employee's own activities. Each item shows the activity name, date, duration in minutes and its Participation Status.

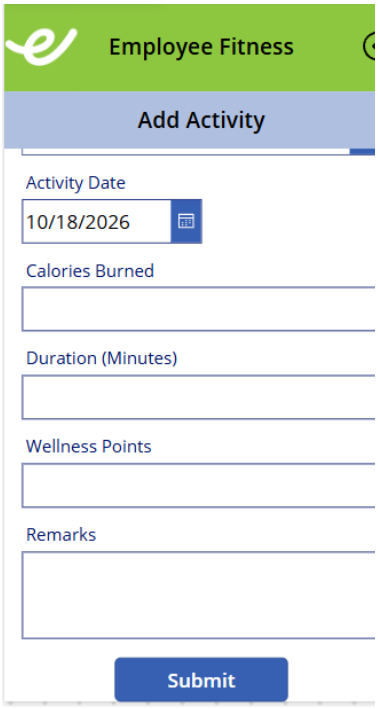
- **Completed / Cancelled** activities are shown for reference only.
- **Pending** activities show a red Cancel button and a green Mark Complete button.
- **Cancel** asks for confirmation before changing the status to Cancelled.



Add Activity (top)

A Power Apps form that creates a new WellnessActivity record in Dataverse. The top of the form contains:

- **Activity Name** - free text.
- **Employee** - shows the signed-in employee's name (lookup).
- **Activity Type** - drop-down from the Dataverse choice column.
- **Activity Date** - date picker, defaulting to today.



Add Activity (bottom)

Scrolling down shows the remaining fields and the Submit button.

- **Calories Burned, Duration (Minutes), Wellness Points** - whole numbers.
- **Remarks** - multi-line notes.
- **Submit** - runs SubmitForm() to save the record. New activities automatically receive the status Pending.



My Progress Screen

Summarises progress for the signed-in employee. Only Completed activities contribute to the totals.

- **Total Activities** 5
- **Completed Activities** 3
- **Wellness Points Earned** 220
- **Total Completed Minutes** 150
- **Calories Burned** 312
- **Progress Percentage** 60 (3 completed out of 5 activities)

6.2 Activity Submission

The employee fills in the form and selects Submit. SubmitForm() writes the record to the WellnessActivity table in Dataverse.

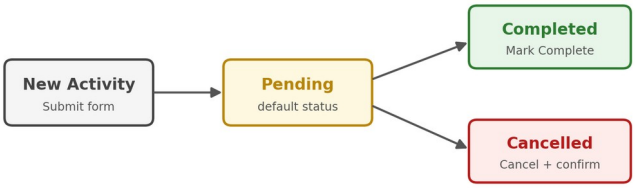


Figure 5: Activity lifecycle

6.3 Activity Lifecycle

- **Pending:** created but not yet performed. This is the default status.
- **Completed:** the employee performed the activity. Completed activities feed the progress calculations.
- **Cancelled:** the employee cancelled the planned activity. Cancelled activities are not counted as completed.

6.4 Cancel and Complete Actions

Cancel: Pending activity, then Cancel, then a confirmation popup, then the user confirms and Participation Status = Cancelled. The popup reduces accidental cancellations.

Complete: Pending activity, then Mark Complete, then Participation Status = Completed, and the activity is included in My Progress.

6.5 Progress Calculation

Metric	How it is calculated
Total Activities	Count of all activities for the employee
Completed Activities	Count of activities with Participation Status = Completed
Wellness Points Earned	Sum of Wellness Points for completed activities
Total Completed Minutes	Sum of Duration (Minutes) for completed activities
Calories Burned	Sum of Calories Burned for completed activities
Progress Percentage	Completed Activities / Total Activities x 100

7. Model-driven App: Views and Forms

Employee Wellness Management Admin is the administrative app used by HR, Wellness Coordinators and Administrators. It provides structured, standard Dataverse forms and views for maintaining wellness data.

7.1 Navigation

The site map contains one group, New Group, with two areas:

- Employee Wellness Profiles - manage employee profiles.
- WellnessActivities - manage wellness activity records.

The command bar offers New, Delete, Refresh, Email a Link, Flow, Run Report, Visualize this view and Share.

7.2 Views

Views let administrators open the records they need without building filters each time.

Active Employee Wellness Profiles

Lists active profiles, sorted by Employee Name. Columns: Employee Name, Email, Department, Designation, Wellness Goal, BMI, Wellness Score and Enrollment Date. The view currently shows six profiles.

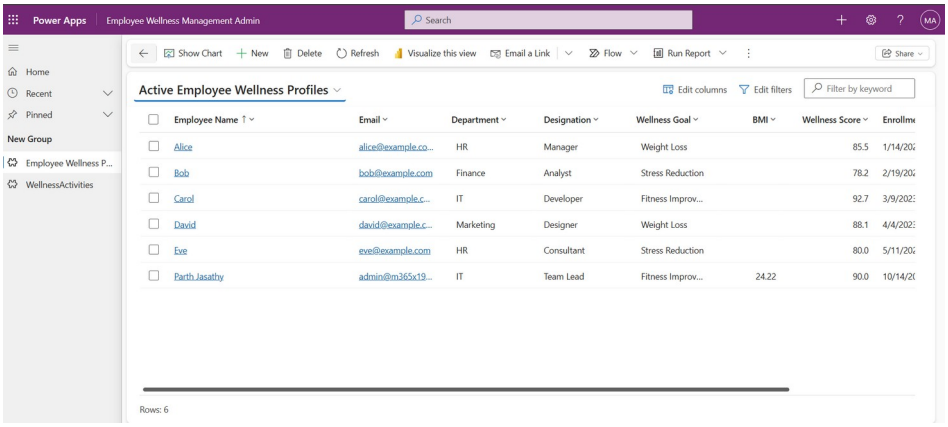


Figure 6: Active Employee Wellness Profiles view

Active WellnessActivities

Lists active activities, sorted by Activity Name. Columns: Activity Name, Activity Type, Activity Date, Duration (Minutes), Calories Burned, Participation Status and Wellness Points.

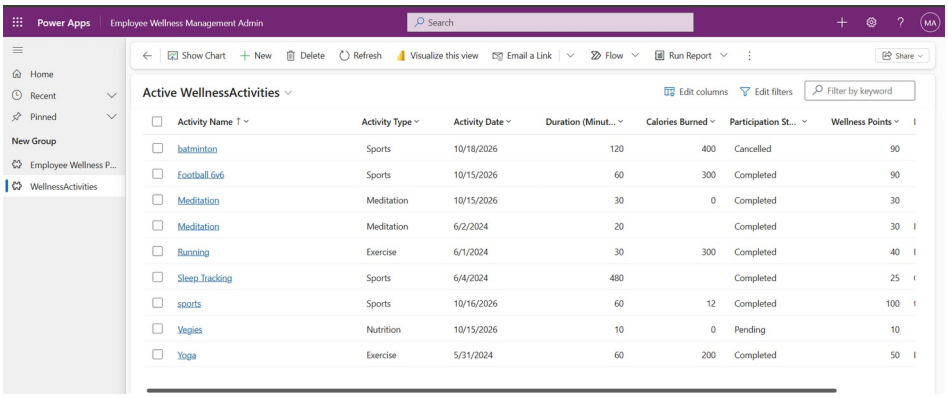


Figure 7: Active WellnessActivities view

7.3 Forms

Employee Wellness Profile form

A main form with two tabs: Personal and Health Details (see Section 10 for the changes made to this form). The Employee Code is shown in the form header.

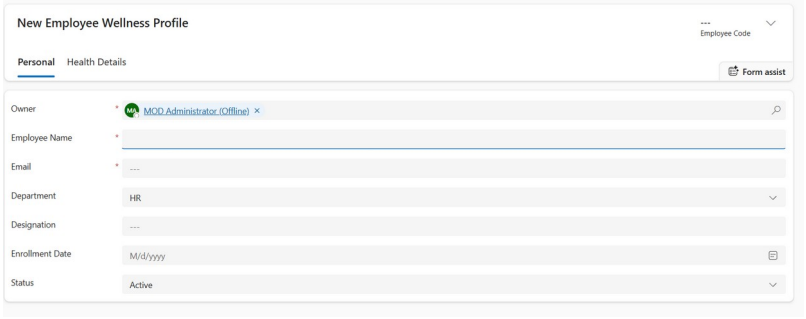


Figure 8: Employee Wellness Profile form - Personal tab



Figure 9: Employee Wellness Profile form - Health Details tab

WellnessActivity form

A main form with a single General tab containing Owner, Activity Type, Activity Date, Duration (Minutes), Calories Burned, Participation Status, Wellness Points and Remarks.

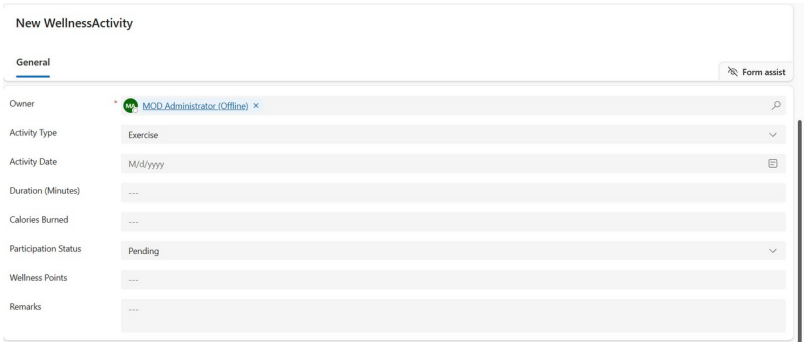


Figure 10: WellnessActivity form - General tab

8. Power Automate Flows

Four cloud flows in the Employee Wellness Solution automate notifications and background processing.

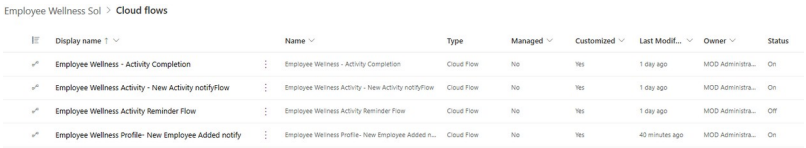


Figure 11: Cloud flows in the Employee Wellness Solution

Flow	Trigger	Purpose	Status
Employee Wellness Activity - New Activity notifyFlow	Dataverse: row added in WellnessActivity	Sends a confirmation email when an activity is created	On
Employee Wellness - Activity Completion	Activity lifecycle (completion)	Handles activity completion and notifies the employee	On
Employee Wellness Activity Reminder Flow	Recurrence (scheduled)	Sends reminders for activities that are still Pending	Off
Employee Wellness Profile - New Employee Added notify	Dataverse: row added in Employee Wellness Profile	Notifies on new profile and sets Enrollment Date to Created On	On

8.1 New Activity Notify Flow

Trigger: Microsoft Dataverse - When a row is added, modified or deleted. Change type: Added. Table: Wellness Activities. Scope: Organization.

Flow structure: New Wellness Activity, then Get Employee (using the Employee lookup's unique identifier), then Send Confirmation.

The employee receives an email containing:

- Greeting with the employee name and a confirmation that the activity was added.
- Activity details: name, type, date, duration, calories burned, wellness points and status (Pending).
- A reminder to open the Employee Wellness app and mark the activity as completed to update progress.

8.2 Activity Completion Flow

Handles the completion stage of the activity lifecycle. When an activity is completed, the flow processes the event and notifies the employee so that the completed activity is acknowledged and reflected in progress.

8.3 Pending Activity Reminder Flow

Runs on a recurring schedule and keeps pending activities visible to employees.

Flow structure: Recurrence, then List Pending Activities, then Apply to each, then Get Employee, then Send Reminder.

The list step filters the activities whose Participation Status is Pending.

This flow is currently turned Off and can be switched on when scheduled reminders are required.

8.4 New Employee Added Flow

Runs when a new Employee Wellness Profile is created. It also populates the Enrollment Date automatically so users never have to enter it by hand:

New Employee Profile, then Power Automate, then Update Employee Profile, then Enrollment Date = Created On.

9. Business Rules

Dataverse business rules enforce data quality on the server side, regardless of whether data is entered through an app, an import or an integration.

9.1 BR - Activity Date Validation

Property	Value
Table	WellnessActivity
Rule name	BR - Activity Date Validation
Condition	Check Act Date: Activity Date is less than Created On
Action	Set Field Value: Set Activity Date to Created On

In text form the rule reads:

```
IF  Activity Date  less than  Created On
THEN  Set Activity Date to Created On
```

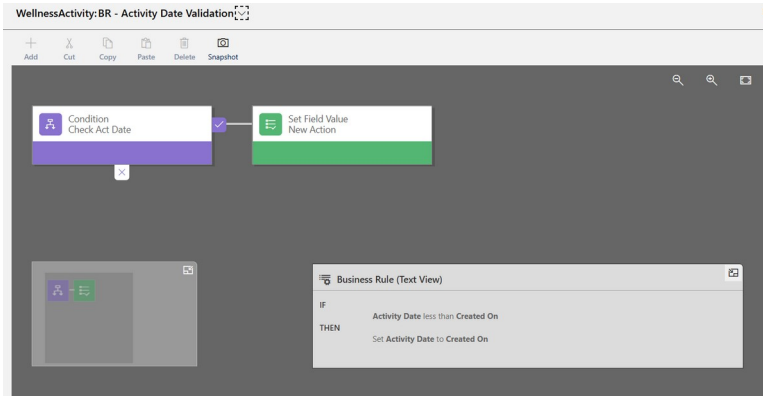


Figure 12: Business rule - Activity Date Validation

Because activities are meant to be scheduled for today or later, a date in the past is corrected automatically instead of being rejected.

Activity Date entered	Result
Earlier than the record creation date	Corrected to the creation date
Today	Accepted
Future date	Accepted

9.2 Other Data Validation

- Employee Name and Email are required on the Employee Wellness Profile.
- Activity Type and Participation Status can only take the predefined choice values.
- Duration (Minutes), Calories Burned and Wellness Points accept whole numbers only.

- Employee is a lookup, so an activity can only be linked to an existing profile.

10. Changes in Model-driven App New Form

The default Dataverse forms were customised to make data entry simpler for HR and Wellness Coordinators.

- New Employee Wellness Profile form: fields are split into two tabs, Personal (owner, name, email, department, designation, enrollment date, status) and Health Details (activity level, height, weight, wellness goal, wellness score). Employee Name and Email are required, Employee Code appears in the header, and sensible defaults such as Status = Active are pre-filled.
- New WellnessActivity form: a single General tab with the activity fields, where Participation Status defaults to Pending to match the Canvas app.

Screenshots are shown in Section 7.3 (Figures 8 to 10).

11. Future Enhancements

- Power BI dashboards covering total, completed, pending and cancelled activities, calories, points and trends.
- Leaderboards, team challenges and reward points.
- Teams notifications, wearable-device integration and role-based dashboards for HR.

12. Outcome

The Employee Wellness Management System replaces scattered spreadsheets with one Dataverse-based solution. Employees can log, complete and cancel wellness activities from a simple mobile app and see their progress instantly, while HR manages profiles and activities through the Model-driven app. Automated flows send confirmations and reminders, and a business rule keeps activity dates valid. The result is better data quality, higher employee engagement and a solid base for Power BI reporting and future wellness features.